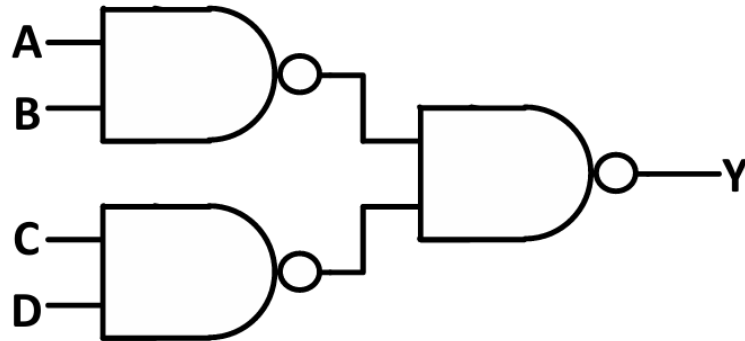


Implementation of the Boolean Function $Y = AB + CD$ Using Raspberry Pi Pico 2 W



1 Problem Analysis

The given logic circuit consists of three NAND gates.

The output of the first NAND gate is

$$X_1 = (AB)'$$

The output of the second NAND gate is

$$X_2 = (CD)'$$

The final output is

$$Y = (X_1 X_2)'$$

Substituting the values of X_1 and X_2 ,

$$Y = ((AB)'(CD)')'$$

Applying De Morgan's theorem,

$$Y = ((AB)')' + ((CD)')'$$

Therefore,

$$\boxed{Y = AB + CD}$$

The output becomes logic '1' whenever:

- $A = 1$ and $B = 1$, or

- $C = 1$ and $D = 1$.

Otherwise, the output remains logic '0'.

2 Hardware Requirements

- Raspberry Pi Pico 2 W Development Board
- USB Type-C Cable
- Personal Computer or Laptop
- LED
- 220 Ω Resistor
- Breadboard and Connecting Wires

3 Pin Connections with Raspberry Pi Pico 2 W

Component	Raspberry Pi Pico 2 W Pin
LED Anode (+)	GP1
LED Cathode (-)	GND through 220 Ω resistor

No external switches are used because the input combinations are generated within the program itself. “latex

4 Hardware Setup and Accessing Raspberry Pi Pico 2 W Using Arduino IDE

The Raspberry Pi Pico 2 W is programmed using the Arduino IDE by installing the necessary board support packages and configuring the development environment. The detailed procedure is as follows.

4.1 Installing Arduino IDE

1. Download and install the latest version of Arduino IDE from the official Arduino website.
2. Open the Arduino IDE after installation.

4.2 Installing Raspberry Pi Pico 2 W Board Package

1. Open Arduino IDE.
2. Navigate to

File $\rightarrow$ Preferences

3. In the “Additional Boards Manager URLs” field, add the following URL:

https://github.com/earlephilhower/arduino-pico/releases/download/global/package_1

4. Click “OK”.
5. Navigate to
Tools → Board → Boards Manager
6. Search for “Raspberry Pi Pico/RP2040”.
7. Install the package developed by Earle F. Philhower.

4.3 Connecting the Raspberry Pi Pico 2 W

1. Connect the Raspberry Pi Pico 2 W to the computer using a USB Type-C cable.
2. If the board is not detected automatically:
 - (a) Press and hold the **BOOTSEL** button.
 - (b) Connect the USB cable while holding the button.
 - (c) Release the button after the board appears as a removable drive.

4.4 Selecting the Board and Port

1. Navigate to
Tools → Board
2. Select
Raspberry Pi Pico 2 W
3. Navigate to
Tools → Port
4. Select the corresponding COM port of the Raspberry Pi Pico 2 W.

4.5 Uploading the Program

1. Write or paste the Arduino program into the Arduino IDE.
2. Click the **Verify** button to compile the code.
3. Click the **Upload** button.
4. Wait until the message
Done Uploading
appears in the output window.

After successful uploading of the program, the Raspberry Pi Pico 2 W executes the Boolean expression

$$Y = AB + CD$$

for all possible input combinations and displays the output through the LED connected to GP1.

5 Implementing Algorithm in Raspberry Pi Pico 2W

5.1 Algorithm

1. Start the program written in the codes sections .
2. Configure GP1 as OUTPUT.
3. Initialize variables A , B , C , and D .
4. Generate all possible combinations of inputs using nested loops.
5. Compute the output

$$Y = (A \& B) | (C \& D)$$

6. If $Y = 1$, turn ON the LED connected to GP1.
7. If $Y = 0$, turn OFF the LED.
8. Wait for one second.
9. Repeat for all sixteen input combinations continuously.

6 Result

The Boolean expression

$$Y = AB + CD$$

was successfully implemented on the Raspberry Pi Pico 2 W using Arduino IDE. The LED connected to GP1 turned ON whenever the output of the logic function was '1' and turned OFF whenever the output was '0'. The program successfully verified all sixteen possible combinations of the four input variables.

